

Technical Questions

1. Could you let us know a few exemplary AI models used for detecting food spoilage, aside from convolutional neural networks?

In addition to Convolutional Neural Networks (CNNs), several AI models can be utilized for food spoilage detection, including:

- Recurrent Neural Networks (RNNs) / LSTM: For sequential data such as time-series sensor readings.
- Random Forests and Gradient Boosting Machines (XGBoost, LightGBM): For tabular or sensor-derived data.
- Support Vector Machines (SVM): For classification based on extracted features like color histograms or texture descriptors.
- K-Nearest Neighbors (KNN): For simpler image or feature-based classification tasks.
- Autoencoders: For anomaly detection in images or sensor readings, e.g., detecting unexpected spoilage patterns.

2. Please let us know on what parameters is the AI model trained on to detect the food spoilage (for example, texture, color, shape etc).

The Teachable Machine-based AI model is trained primarily on visual cues from food images, including:

- Color changes (e.g., discoloration or browning).
- Texture alterations (e.g., shriveling, surface degradation).
- Shape deformations (e.g., swelling or collapse).
- Pattern changes (e.g., mold spots or irregular markings).

These parameters enable the model to classify foods into Fresh, Moderately Fresh, or Spoiled.

3. The IDF mentions that “Another unique aspect is the intelligent volunteer assignment module, which uses live location tracking and route optimization to reduce pickup delays and improve delivery efficiency”. Please let us know if the invention provides any unique solution on way the route optimization is performed or volunteer is assigned intelligently, or are the existing route optimization processes used?

The invention leverages existing route optimization methods using Leaflet with OSRM (Open Source Routing Machine) for shortest-distance routing between coordinates. OSRM’s routing engine uses Dijkstra’s algorithm + A* (with heuristics) + Contraction Hierarchies to find the shortest/fastest path efficiently. While this module improves operational efficiency, it does not introduce a novel algorithm for route optimization or volunteer assignment; it relies on live location tracking combined with standard routing solutions to reduce pickup delays.

4. We understand that there is a handheld device with different sensors used to detect various parameters such as temperature, moisture, gases etc, and various levels are set to detect the freshness of food. We believe that these levels vary based on the type of food being tested. In such a case, how are these levels set for use of this device with different varieties of food?

The proposed handheld device employs a **dual-switch operational mechanism** to enable accurate food freshness detection across multiple food categories without manual category selection. Upon pressing the first switch, the integrated ESP32-CAM captures an image of the food sample and transmits it to a server or processing unit, where image analysis is performed to determine the type of food (such as vegetables, grains, or cooked food). Based on the identified category, the system automatically updates the threshold profiles corresponding to spoilage-related parameters. When the second switch is pressed, the device activates the MQ-135 gas sensor and the DHT22 temperature-humidity sensor to capture real-time data. These sensor readings are then compared against the dynamically updated thresholds to classify the food as Fresh, Moderately Spoiled, or Spoiled. This two-step approach allows for category-specific, one-shot freshness detection without requiring repeated measurements or manual category input, thereby enhancing both usability and accuracy.

5. We would like to have more data on how the expiry date of food is estimated in the invention. Particularly, what are the parameters considered? It would be helpful if you can provide an example here.

In the hybrid system, expiry estimation is performed using both image-based AI detection and linear regression modeling:

- Image-based AI model predicts the freshness category (Fresh, Moderately Fresh, Spoiled) based on visual features such as color, texture, and pattern.
- Linear regression model estimates the remaining shelf life by correlating the predicted freshness with historical spoilage data, environmental conditions, and storage parameters.

Example for tomatoes:

- Fresh: 3–5 days remaining → system predicts an average of 4 days.
- Moderately Fresh: 1–2 days remaining → system predicts 1.5 days.
- Spoiled: 0 days → considered expired.
The expiry date is calculated by adding the predicted remaining shelf life (based on freshness category and regression output) to the current date. This hybrid approach improves accuracy over using image or sensor data alone.

Questions on Inventive Step

1. A key part of the patentability assessment is the "inventive step" requirement. The patent office tests this by asking whether the invention would be obvious to a person with your same skill set, either alone or by combining information from existing publications. For this, we have included the link of the publication which we discussed during the call below for your reference:

<https://www.frontiersin.org/journals/public-health/articles/10.3389/fpubh.2021.816226/full>

Please provide us the differences of the invention w.r.t the document in above link:

- The publication uses sensor-based spoilage detection (gas, humidity, temperature) combined with CNNs for classification.
- Our invention uses ESP32-CAM + Teachable Machine AI for image-based, non-contact detection, eliminating the need for multiple sensors and enabling portability.
- Multi-class freshness classification (Fresh, Moderately Fresh, Spoiled) allows more granular assessment compared to the binary safe/unsafe classification in the publication.
- The system can be retrained for different food types without hardware changes, unlike sensor-based systems that require recalibration.
- While the referenced publication utilizes a Raspberry Pi 5, our system employs an ESP32 to achieve a cost-effective and budget-friendly solution without compromising functionality.

System Architecture – Dual Modules:

1. Donor Module (App-Based Image Upload):

- Donors can upload images of food via a smartphone application.
- Uses an image-based AI model (Teachable Machine) to analyze color, texture, and pattern.
- Provides real-time predictions of freshness and expiry estimation using a hybrid AI + linear regression approach.
- No sensors are required, making it accessible and easy to use.

2. NGO Module (Handheld Device):

- NGOs use a handheld device equipped with sensors (DHT22, MQ135) and ESP32-CAM.
- Integrates sensor readings with AI-based image analysis to monitor inventory.
- Hybrid model ensures accurate expiry predictions across various food types.

Technical Advancement Over Prior Art:

- Unlike the publication, which primarily uses sensors or a single approach, our system combines **image-based AI for donors** and **sensor + AI handheld monitoring for NGOs**, providing flexible, real-time predictions and scalable inventory management.
- Significant engineering was required to integrate image processing, sensor fusion, AI model deployment, and user interface, demonstrating a non-obvious inventive step.
- Training on images for donors enables feature learning like color, texture, and pattern, providing scalability and adaptability to new food types, representing a technical advancement over the sensor-based systems described in the publication.